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Descartes' Notion of the Mind-Body Union and its Phenomenological Expositions¹

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Neuroscientists, neurophilosophers, and experts in cognitive science often present their own research programs in opposition to a doctrine called “Cartesian dualism.”

According to them, this doctrine consists in the radical ontological separation of the mental and the physical, mind and body. The neurologist António Dámasio is well known for such claims. In *Descartes' Error* (1994), Dámasio summarizes the Cartesian misconception as follows:

This is Descartes' error: the abyssal separation between body and mind, between sizable, dimensioned, mechanically operated infinitely divisible body stuff, on the one hand, and the unsizable, undimensional, un-pushpullable, nondivisible mind-stuff; the suggestion that reasoning, and moral judgment, and the suffering that comes from physical pain or emotional upheaval might exist separately from the body. Specifically: the separation of the most refined operations of the mind from

¹ We are grateful to Professor Lilli Alanen for clarifying discussions on Descartes' metaphysics, epistemology, and philosophy of passions, and to the two anonymous readers who commented on an earlier version of this paper and offered valuable suggestions for revision.

the structure and operation of a biological organism. (Damasio 1994: 249–250; cf. Carman 2008: 15–27, 53–54)

As an alternative to “Cartesian dualism,” Damasio proposes a neuroscientific account according to which the cognitive, emotive, practical, and discursive operations characteristic of human life, emerge from and supervene on biological processes of neural signaling and cerebral decoding.²

Whatever the value of such neuroscientist and neurophilosophical theorizing may be – and we believe it to be considerable – it is philosophically crucial to see clearly how this theorization relates to the philosophy of René Descartes that it is claimed to oppose or replace.

Neuroscientific and neurophilosophical critiques of “Cartesian dualism” tend to emphasize one element of Descartes’ philosophy but neglect another central factor. To be sure, Descartes made a clear *conceptual* distinction between the properties of mind and those of matter – thought and extension –, but he also argued that mind and body are *in fact* connected.³ By arguing that our mental acts, for example acts of willing and deciding, are factually always accompanied by material events in the brain, and that

² Cf. Petitot, Varela, Pachoud, and Roy (eds.) 1999; Varela, Thompson, and Rosch (eds.) 1991; Maturana and Varela [1987] 1992; Maturana and Varela [1972] 1980. Many contemporary philosophers who develop theories of autonomous biological organization of life or *autopoiesis* refer to Merleau-Ponty’s discussion of perception and further to the work of the Estonian-German biologist Jakob von Uexküll ([1934] 2004). For Merleau-Ponty’s discussion of Uexküll’s conceptualization of the organism-environment relation, see his late lectures on *Nature* (1995).

³ Cf. Merleau-Ponty [1945] 1993: 111–112/82–83.

mental events, for example desires and affects, are systematically connected with cerebral traces (Descartes AT XI: 407, 428; CSM I: 365, 375–376; letter to Chanut, February 1st 1647, AT IV: 600–617; CSM-K: 305–314; cf. Husserl 1952a: 288–297/302–311⁴), Descartes established a new research paradigm that later grew into the comprehensive program of modern psychophysiology and also includes contemporary inquiries into the cerebral-neural basis of mental processes. So, rather than being rejected as a metaphysical or ontological dualist, Descartes could well be celebrated as the forefather of contemporary neuroscience and neurophilosophy with their hypotheses of connectionism, embodied cognition, enactivism, and extended mind (e.g. Hutto and Myin 2013; Rietveld 2013; Noë 2004; Clark 1997).

But Descartes' hypothesis has even more far-reaching effects: his insight into the systematic correlation between mental events and physiological events, explainable by purely mechanical laws, has paradoxically also allowed the construction of purely materialist theories of the mind. His correlationistic and interactionistic insights made conceptual and theoretical space for later attempts to build materialist theories of the mind that identify the mental and the neural and purport to explain mental causation and motivation on the basis of physiological mechanisms (cf. Kaitaro 2007; 2004). These alternatives range from the classical non-reductionistic materialism of La Mettrie and Diderot (cf. Kaitaro forthcoming) to twentieth-century reductionistic and eliminativistic approaches and further to the supervenience theories and token-identity theories

⁴ The pagination given first refers to the original source, and the pagination that follows this, after the slash, refers to the English translation. Both sources are given in one and the same entry in the list of references.

developed by twentieth-century philosophers of mind (e.g. Davidson 1995, 266; Davidson 1993; Churchland P.M. 1988; Churchland P.S. 1986; Davidson [1970] 1980).

However, it is also crucial to realize that while insisting on the conceptual distinction between the mental and the material, Descartes created conceptual space for the study of mental phenomena as such, that is, as experienced actions and reactions with temporal and intentional structures identifiable and describable independently of their material underpinnings. At the turn of the twentieth century, this possibility was systematically explored by Franz Brentano and his followers under the titles “descriptive psychology” and “eidetic psychology.”

These two aspects of “Cartesian dualism” – the possibility to characterize the mental as such and the possibility to find systematic connections between the mental and the material – are in fact correlative and complementary aspects of Descartes’ account of the human mind: they are both consequences of the conceptual distinction between mind as thought and body as extension.

Later these two approaches to the mental have been developed independently from one another and also associated with opposite metaphysical and scientific aspirations, materialism vs. idealism, objectivism vs. subjectivism, and scientism vs. humanism. In the twentieth century, we have learned to take them as alternative explanatory paradigms: scientists and science-inspired philosophers have been critical of the study of the human mind as an essentially intentional and teleological system, and mentalistic philosophers have been critical of reductionistic, materialistic, and causalistic theories of human cognition. As the French phenomenologist Maurice Merleau-Ponty so aptly

put it in his late essay, “Eye and mind” (*L’Œil et l’esprit* 1964): “Our science and our philosophy are two faithful and unfaithful offshoots of Cartesianism, two monsters born of its dismemberment” (Merleau-Ponty 1964a: 58/138; cf. 1964b: 17/11, 242/188, 251/198, 288/234).

1. Pure Mind and the Mind-Body Union

Descartes himself was able to combine the two approaches to the mental – mind studied as such and mind studied in relation to the body – since he distinguished between three different ways of knowing and three proper objects of human knowledge. He argued that our intellection or understanding (*entendement*), when operating alone, provides us with sufficient knowledge of pure mind or mind as the thinking substance, whereas we should use both our understanding and our imagination in order to have knowledge of the nature of the extended substance or pure matter.⁵ Thus, the faculties of understanding and imagination guaranteed our knowledge of the two substances. But in addition to them, Descartes argued, our knowledge also includes familiarity with the union of mind and body, and the knowledge of this union depends on our sensory faculties:

The soul is conceived only by the pure intellect; body (i.e. extension, its shapes

⁵ For Descartes, imagination means considering something that is not really there “as if it were present” to oneself (Descartes AT VII: 72; CSM II: 50). In distinction from understanding (*entendement*), imagination is a hybrid faculty of knowing in involving both mental and bodily components. This implies that purely mental beings (e.g. angels) cannot have the faculty of imagination, and thus this faculty is specific to the human mind. In distinction from humans, God and his angels proceed by pure intellect and will in all their thoughts.

and motions) can likewise be known by the intellect alone; but much better by the intellect aided by imagination; and finally what belongs to the union of soul and body is known only obscurely by the intellect alone or even by the intellect aided by the imagination, but it is known very clearly by the senses. (Descartes' letter to Elisabeth, June 28th 1643, AT III: 691–692; CSM-K: 227)

So, according to Descartes, the three objects of human knowledge are: first, mind as pure thought, second, body as extension, and third, the mind-body union. These are discovered and known by us thanks to three different mental faculties. Pure mind is known by understanding alone, and the extended body is best known by understanding aided by imagination. But to know the soul-body union, Descartes argues, we need to interrupt our metaphysical meditations and arrest our aptitude for calculation and imagining, and pay attention to our sensations and to our interpersonal relations.⁶ In a letter to his pupil Princess Elisabeth of Bohemia, Descartes explains:

Metaphysical thoughts, which exercise the pure intellect, help to familiarize us with the notion of the soul; and the study of mathematics, which exercises mainly

⁶ Thus, Descartes argues that while studying the extended substance we can either use our intellect independently of our imagination, imitating the thoughts of angels, or else boost the operations of our intellect with our imagination. This seems to be in tension with the explicit teaching of *Meditations*, where Descartes argues that imagination is not trustworthy as a means of representation but is deceptive in producing hallucinations and delusions. However, *Meditations* also demonstrates that free imagination has an important role in the development of human knowledge. It cannot provide certainties but it is indispensable in the production of possibilities, and some of its creations – such as the possibility of a malicious demon, summoned up by Descartes himself – promote our knowledge and deepen our understanding of its grounds (cf. Sepper 1996). Consideration of extreme possibilities is thus not merely an entertainment but is crucial to the sciences of metaphysics and mathematics.

imagination in the consideration of shapes and motions, accustoms us to form very distinct notions of body. But it is the ordinary course of life and conversation, and abstention from meditations and from the study of things which exercise imagination, that teaches us how to conceive the union of the soul and the body.

(Descartes' letter to Elisabeth, June 28th 1643, AT III: 692; CSM-K: 227)

In his correspondence with Elisabeth, Descartes modifies and complements the dualistic account of his *Meditations on First Philosophy* by a discussion of the mind-body union. He calls this union “the third primitive notion” and argues that it is not reducible to pure mind nor to pure matter (Descartes' letter to Elisabeth, May 21st 1643, AT III: 663–668; CSM-K: 217–220; letter to Elisabeth, June 28th 1643, AT III: 690–695; CSM-K: 226–229). Further Descartes claims that it is this notion of union, and not the two notions of separate substances, that allows us to conceive how our will can move our body and how external events and entities affect our sense organs and our sensibility.

Many commentators neglect the methodological implications of Descartes' discussion of the mind-body union, and present this discussion as a mere therapeutic or pedagogic supplement to his metaphysical system, intended to reconcile the insistent Princess and to advise her in her practical predicaments.⁷ But Descartes is coherent in approaching the problem of the union in an epistemologically controlled manner. When he explains to Elisabeth that there are in fact three and not just two primitive notions, he insists on the fact that each of these can be known in a particular way and should not be studied in

⁷ One scholar who has systematically emphasized the importance of Descartes' exchange with Elisabeth in the development of his philosophical system, is Lisa Saphiro (2013, 1999). See also Saphiro and Pickavé (ed.) 2012; Saphiro (ed.) 2007; Alanen 2003; Tollefsen 1999.

comparisons with, or by reduction to, any other. Matters pertaining to the union can only be known obscurely by the understanding or intellection. Resorting to imagination does not make this notion any clearer or distincter either. What one has to realize is that the notion of the union can be known very distinctly only by the senses (Descartes' letter to Elisabeth, June 28th 1643, AT III: 691–692; CSM-K: 226–227).⁸

It is philosophically crucial to notice that the idea of conceptual distinctness that Descartes develops in *Meditations* applies not only to the notions of pure mind and pure matter, but also to the third notion, the notion of union. This implies that one should not explain the way the mind interacts with the body by the conceptual tools that are used when explaining the mutual interactions of bodies. Nor can one understand the union by merely applying concepts that are developed for the correct description of pure mind and its principal attribute of thinking. Ultimately, the result of Descartes' exchange with Elisabeth can be summed up by the epistemological and methodological thesis that the mind-body union must not be approached by the metaphysical and physical concepts developed in his *Meditations*.

The difficulty that we experience in trying to conceive of the union of mind and body results, according to Descartes, from our incapacity to distinctly conceive at the same time the three different notions: mind as pure thought, body as extension, and the union of mind and body (Descartes' letter to Elisabeth, June 28th 1643, AT III: 693; CSM-K: 227–228). In order to relieve this difficulty, Descartes suggests to Elisabeth that she should deviate from the reflective order of the *Meditation* and should feel free to attribute extension to the mind. As the Princess had remarked, this would be somewhat

⁸ Cf. Heinämaa 2003; Garber and Wilson 1998: 835.

easier than to conceive a completely non-extensional mind as moving and as being moved by material things.

In this context, Descartes observes that attributing extension to the mind is actually identical with conceiving its union with the body. However, he feels it necessary to add not only that the extension that is attributed to thought in this manner is distinct from thought as such (*la pensée même*), but also that the extension of thought is different from the extension attributed to the body as mindless being. When extension is attributed to pure matter it is a determination of a certain place that excludes all other extensions, but this is not the case for the kind of situational extension that we need to attribute to the mind in order to conceive its union with the body (Descartes' letter to Elisabeth, June 28th 1643, AT III: 694; CSM-K: 228).

So, according to Descartes, there are actually two facts on which our knowledge of the nature of the mind depends: first, that the mind thinks, and second, that it is united with the body and acts and is acted on by it.⁹ These correlate with two different conceptions of extension, extension as conceived in a system of coordinates and extension as situation (e.g., Descartes AT IXB: 47–48; CSM I: 228–229; cf. Casey 2013: 135–137, 151–152, 158–159). Descartes main purpose in the *Meditations* was to demonstrate the

⁹ In a letter to Regius (January 1642), Descartes explains: “When we said that a human being is an *ens per accidens*, we meant this only in relation to its parts, the soul and the body; we meant that for each of these parts it is in a manner accidental for it to be joined to the other, because each can subsist apart, and what can be present or absent without the subject ceasing to exist is called an accident. But if a human being is considered in himself as a whole, we can of course say that he is a single *ens per se*, and not *per accidens*; because the union which joins a human body and soul to each other is not accidental to a human being, but essential, since a human being without it is not a human being” (AT III: 508; CSM-K: 209).

conceptual distinction of mind from body, and thus he did not discuss the connectedness of mind with matter but merely mentioned the phenomenon in the sixth Meditation (Descartes' letter to Elisabeth, May 21st 1643, AT III: 664–665; CSM-K: 217–218). Consequently, there is, according to Descartes, a sphere of human knowledge which remains ambiguous to understanding and which we cannot know by the method that he devises for metaphysical inquiries.

Descartes struggles to conceptualize this ambiguous sphere of the union in correspondence with his pupils and opponents while discussing the alleged problems of his *Meditations*. Later, in *The Passions of the Soul*, he tackles the problem explicitly, trying to develop a coherent account that would do justice to the notion of the union as well as the idea of conceptual distinctness. Here he distinguishes between three types of mental phenomena that all belong to the ambiguous sphere of the mind-body union: sensations, sense perceptions, and emotions. In order to account for these phenomena, he develops a twofold description that characterizes mental states as they appear to the persons who experience them and at the same time as they can be captured in terms of the developing sciences of anatomy and physiology. Thus, he provides an account of both the representational-intentional and the causal features of sensations, sense perceptions, and emotions. On the one hand, he describes how these states (re)present their objects, specifying, for example, what value and what temporality an emotion ascribes to its objects. On the other hand, he describes the kinds of material (cerebral, neural, haematic¹⁰) movements that relate to such states as their causes and effects. Joy, for example, is defined by Descartes as “the soul's enjoyment of a good that impressions in the brain represent to it as its own” (Descartes AT II: 396; CSM I: 360),

¹⁰ “Haematic” means “relating to the blood.”

and desire is defined as “an agitation of the soul caused by spirits that disposes the soul to want to have in the future the things it represents to itself as agreeable” (Descartes AT II: 392; CSM I: 358; cf. Heinämaa forthcoming). Both definitions operate with both representational-intentional and causal concepts. The implications of this explanatory twofoldness are equally important to our contemporary theorization as are the implications of Descartes’ metaphysical theory of the two substances.

2. Interactionism and Holism

It is crucial to notice that Descartes develops his double account of the mind-body union by means of two central metaphors.

On the one hand, he describes the mind-body relation with the concepts of interaction and mutual causal influence. The ideas of interaction and causation suggest that some part of the body functions as the principal seat of mental operation. Since organs of sensation are known to be connected to the brain by neural transmission, it seems natural to assume that the brain is the organ that primarily interacts with the mind, receiving from it the directions of the will and transmitting to it the impulses from the sense organs. However, Descartes argues even more ambitiously that our mental control of our bodies is further centralized and that a particular cerebral gland, the pineal gland, serves as the principal bodily locus in which the human mind directly “exercises its functions”:

[T]he soul has its principal seat in the little gland which exists in the middle of the brain, from whence it radiates forth through all the remainder of the body by

means of the animal spirits, nerves, and even the blood. (AT XI: 354; CSM I: 341)¹¹

On the other hand, Descartes insists that the mind-body union is an inseparable whole of two principles that are comprehensively and completely “intermingled”: “The whole mind is united to the whole body,” he declares in a letter to his pupil Regius (Descartes’ letter to Regius, January 1642, AT III: 493, cf. 508–509; CSM-K: 206, cf. 209).

Moreover, in both *Meditations* and *Discourse*, he contrasts the holistic metaphors of intermingling explicitly with the traditional Platonic images of piloting and control, and argues that we cannot make sense of our sensations unless we let go of the imagery of mental piloting. In the sixth Meditation, we read:

Nature also teaches me, by these sensations of pain, hunger, thirst and so on, that I am not merely present in my body as a sailor is present in a ship, but that I am very closely joined and, as it were, intermingled with it, so that I and the body form a unit. If this were not be so, I, who am nothing but a thinking thing, would not feel pain when the body is hurt, but would perceive the damage purely by his intellect, just as a sailor perceives by sight if anything in his ship is broken.

(Descartes AT VII: 81–82; CSM II: 56)

¹¹ Descartes discusses the function of the pineal gland in *The Passions of the Soul* and in *Treatise of Man*, which was his first book (written before 1637, but published posthumously, first in Latin translation in 1662, and then in the original French in 1664). The topic is also debated in a number of letters written in 1640–1641. For a detailed account of these discussions and their background in Greek, Medieval, and Renaissance medicine, see Lokhorst 2016; Lokhorst and Kaitaro 2001.

Thus, our philosophical alternatives are either (i) to identify ourselves with the intellect and free will, and fail to make philosophical sense of our sensations, or else (ii) to allow the concept of thought to cover – in its impure modes – not just the actions of the intellect and the will, but also the passions of sensation, sense perception, and emotion.¹² Descartes consistently opted for the second alternative. For him, the human mind is not merely composed of mental *acts* but also includes *passions*. In *Discourse on Method*, he summarizes his position as follows:

I showed how it is not sufficient for [the mind] to be lodged in the human body like a helmsman in his ship (...) but that it must be more closely joined and united with the body in order to have (...) feelings and appetites like ours and so constitutes a real man. (Descartes AT VI: 59; CSM I: 141)

The metaphor of radiation used in *Passions* allows Descartes to develop this holistic intuition while at the same time entertaining the idea of an operative center in the brain. However, his characterization of sensations and sense perceptions as intrinsically mental suggests a stronger holistic view according to which sensations are not just the ultimate frontier between thought and extension but are themselves both mental and extended in the special situational sense implied by the notion of the mind-body union. This indicates that the human mind is dispersed throughout the whole human body, which undermines the hypothesis of a distinguishable bodily seat of mental activity.

¹² Thus, the mind for Descartes is not merely pure thought – acts of the will and the intellect – but includes also the passions and imagination.

In many commentaries, the twofoldness of Descartes' description of the mind-body union is explained away by suggesting that the holistic notion of the union is either a non-philosophical courtesy to the Catholic Church or else an Aristotelian relic (e.g. Garber and Wilson 1998: 834–836). Thus, Descartes' only proper thesis about the mind-body relation would be the thesis about centralized interaction (with all the problems related to interactionism).

However, in *Phenomenology of Perception*, Merleau-Ponty puts forward an alternative reading by arguing that Descartes' twofold account of embodiment anticipates later phenomenological explications of bodily forms of intentionality, originally developed by Edmund Husserl and further elaborated by himself. More precisely, Merleau-Ponty maintains that we can make sense of the twofoldness of Descartes' account if we utilize phenomenological tools in our analysis of bodily experiences and experiences of bodies.

When Descartes says that the understanding knows itself incapable of knowing the union of soul and body and leaves this knowledge for life to achieve, this means that the act of understanding presents itself as reflection on an unreflective experience which it does not absorb either in fact or in theory. (Merleau-Ponty [1945] 1993: 52–53/37)

Merleau-Ponty's *Phenomenology of Perception* thus argues that Descartes' idea of sense perception and sensation as intrinsically mental faculties can be explicated and refined by Husserl's phenomenological account of perceptual consciousness and motor intentionality. Moreover, since Husserlian phenomenology also includes an argument for the primacy of perception, i.e., an argument to the effect that perception provides a

necessary foundation for conceptualization, its account of perception has important implications for our epistemological theories of conceptual knowledge (e.g. Merleau-Ponty [1945] 1993: xiii/xvi–xvii, 418–419/326–327). Eventually Merleau-Ponty’s discussion of Descartes suggests that the tension between Descartes’ two descriptions of the mind-body union – the interactionistic and the holistic – can be alleviated or mitigated by Husserl’s distinction between different ways of conceptualizing the world and worldly relations, all veracious but valid on different conditions.

3. Two-Layered Complexes or Expressive Wholes?

In the second volume of his mature work *Ideas*, Husserl distinguishes between two different attitudes and two different interests in which we can study living beings, humans and animals. On the one hand, we may want to construe a theory of such beings in order to explain and predict their behaviors, forms of behavior, or physical constitutions. On the other hand, we may need to communicate with such beings and inform them about our beliefs, emotions, aims, and needs, and correspondingly interpret their analogous intentions.

In the first case, we tend to proceed in the naturalistic attitude that conceives living beings as psychophysical systems in complex causal-functional relations to the environment and enviroing entities. In the second case, we usually adopt the personalistic attitude that conceives living beings as autonomous agents in intricate communicative relations with other similar beings. When we proceed in the naturalistic manner, the positions and movements of the living bodies that we study are conceptualized by us as various effects of external and internal causes, and we may

want to manipulate these causes in order to produce beneficial effects. When we proceed in the personalistic manner, the very same positions and movements are understood as communicative gestures that require our responses and have sense – or occasionally lack sense – as parts of significative systems (Husserl 1952a: 183–239/192–250; cf. Stein 1917: 44–60/41–54).

According to the naturalistic paradigm, each being is either itself physical, belonging to the unified totality of physical nature, or a variable dependent on the physical, and thus at best a secondary accompaniment or adjunct. The human person is conceived as a two-layered reality, a compound system, in which a material – biological, biochemical, chemical, and physical – basis provides the foundation for the emergence of psychic or mental features. Thus understood, psychic or mental features are not properties of any types of immaterial entities – souls, agents or persons – but are higher-order properties emergent of immensely complex physical systems, e.g. the brain or the neural network.

Husserl lays great stress on the experiential fact that we do not always, or even predominantly, grasp human bodies as two-layered unities. Rather than characterizing our everyday concern for our fellow beings, this attitude is characteristic of the scientist's explanatory interest. By default, we attend to other living beings, both humans and animals, as unitary expressive and intentional agents, and this is also our basic attitude when we encounter a novel type of being that moves in the spontaneous and reflective manner, characteristic of the living, such as a robot (cf. Husserl 1952a: 151–152/159).

Moreover, our most central human practices require that we relate to one another as expressive beings and not as complicated bio-chemical mechanisms. These include, not just the emotively charged practices of caring and nurturing, but also the practices of arguing, criticizing, experimenting, and testing, which are all central to the scientific enterprise itself. In order to guarantee the validity of our results, we need to subject our hypotheses, our inferences, our experimental designs, and our conceptual distinctions to the critique of other scientists, and in this essential task we cannot proceed by maintaining that our fellow beings are complicated machines manipulable by us and others. Thus, the personalistic attitude, which gives the living being as an expressive whole, is necessary both practically and epistemically (e.g. Husserl 1952a, 204ff./214ff).

Husserl clarifies the givenness of living beings by comparing it to that of literary works (Husserl 1952a: 236ff./248ff.). The main point of the comparison is to draw attention to the way in which such entities are articulated and structured in experience. He argues that neither persons nor cultural works can be analyzed into parts in the same manner as physical and psychophysical systems can (Husserl 1952a: 134/141). Rather than having mental contents or sense contents as layers founded on a non-significative substratum, these unities are completely permeated by sense. Each part and each layer that can be disclosed by analysis is filled with sense; and no non-significative ground can be detected by analysis. Moreover, each expressive totality, and each part of such a totality, has multiple relations of sense to other expressive units (Husserl 1952a: 252–253/241, cf. 132–139/139–147).¹³

¹³ See also Merleau-Ponty [1945] 1993: 211–212/162–163.

Thus understood, the expressive gestures of persons have a sensible substratum, but the substratum in this case is not a physical thing or a merely material object. The substratum of the smile, or a grimace, is the face that always and already is signifying and never simply there at hand as physical things are (Husserl 1952a: 239/251, cf. 246/234–235).¹⁴

The face and the expressive body belong to the personalistic world of communicative and motivational relations. In it, humans act as “ensouled” bodies, relating to one another and to their environments by signifying bodily movements and ways of moving (Husserl 1952a: 234ff./ 246ff, 285–286/298–299). Via such bodies, the whole environment is articulated into a system of meaningful entities that call for our attention and move us by their powers of attraction and repulsion. In *Phenomenology of Perception*, Merleau-Ponty sums up this line of thought as follows:

This disclosure of an immanent or incipient significance in the living body extends, as we shall see, to the whole sensible world, and our gaze, prompted by the experience of our own body, will discover in all other “objects” the miracle of expression. (Merleau-Ponty [1945] 1993: 230/177)

The naturalistic and the personalistic apprehensions of the living body are not on equal footing according to Husserl’s analysis. He argues that the primary way in which living bodies are given to us is personalistic and as such expressive or significative: “The question above was: is a human being [*Mensch*] a unification of two realities, is that

¹⁴ See also Merleau-Ponty [1945] 1993: 175–176/133–134, 186–187/142, 193/147–148, 215–217/165–167, 271/210.

how I see him? If I do, then I am grasping a corporeal existence. But that is not the attitude I am in when I grasp a human being. I see a human being, and in seeing him I also see his body” (Husserl 1952a: 240/251–252). We do not *perceive* living bodies as bio-mechanisms but as expressive gestures, motivated by sensations and feelings, directed by intentions and purposes, and responsive to affects and appeals. And it is perception, not understanding or pure thinking, that gives us living beings *per se*.

In this way, Husserl questions the alleged primacy of the naturalistic theoretical attitude and argues that this attitude depends on a more profound perceptual stance in which we do not study human beings, ourselves and others, as psychophysical complexes but experience and conceive them as unitary persons.¹⁵ Persons are not two-layered realities with causal coherence but unified phenomena of expression. As such, the human mind is not an adjunct to an independently operating material body, but is the principle or norm according to which matter is organized into significative wholes in significative relations.

Thus, Husserlian phenomenology allows us to distinguish between two different ways of studying the mental, either as a psychophysical complex in which the psychic is founded upon the physical, or else as an expressive whole in which matter is articulated by sense. This gives a more precise formulation to Descartes’ twofold account of the mind-body union, interactionistic on the one hand, and holistic on the other. But unlike Descartes, Husserl also provides an explication of the relations of these two ways of

¹⁵ Ultimately, Husserl’s thesis about the dependency of the naturalistic attitude on the personalistic attitude rests on his theory of individuation. For a discussion of this background, see the chapter on “Embodiment and Bodily Becoming” included in this volume.

apprehending the mind-body union. He argues that both approaches are legitimate within their own limits but that the naturalistic approach is dependent on the personalistic one in the task of individuating its research objects.

4. On the Tasks of the Science of Philosophy

Husserl separates both these approaches to the human mind from his own transcendental-phenomenological inquiries that are designed to illuminate the activities of sense constitution and thus provide the foundations for both the other approaches. He argues that it is necessary to radicalize philosophical reflections in order to delineate the proper objects of the positive sciences, and their differently interested attitudes, and thus to clarify their relations of dependence and interdependence.

In his mature works, *Cartesian Meditations* (*Cartesianische Meditationen* 1929, 1931) and *The Crisis of European Sciences and Transcendental Phenomenology* (*Die Krisis der europäischen Wissenschaften und die transzendente Phänomenologie* 1936–1937, 1954), Husserl makes clear that the main motivation of his philosophy coheres with that of Descartes: the aim is to provide rational foundations for all the sciences and for all the practical endeavors of human beings. In *Cartesian Meditation*, he characterizes his project as follows:

Must not the only fruitful renaissance be the one that reawakens the impulse of the *Cartesian Meditations*: not to adopt their content but, in *not* doing so, to renew their spirit, the radicalness of self-responsibility, to make that radicalness true for the first time by enhancing it to the last degree, / to uncover thereby for the first

time the genuine sense of the necessary regress to the ego, and consequently to overcome the hidden but already felt naïveté of earlier philosophizing (Husserl 1950: 47–48/6; cf. 1952b: 141/408).

Husserl's first emphasis is on the fact that Descartes' *Meditations* developed in an innovative new way the idea of philosophy as a universal science. Descartes saw the importance of the unity of science and realized that this unity is not thematic or methodic but rests on the sameness of tasks and goals: each specific science – from metaphysics and geometry to quantum physics and from poetics to neuroscience – aims at providing an evidence-grounded conception of its object (Husserl 1950: 50ff/9ff.; 1954: 75–82/302–303). Thus, the ideas of grounding, justification, and evidence belong to the common core of all sciences. Philosophy is a foundational science in investigating the conditions of the different types of evidences by which the other sciences operate. In this context, Husserl even contends that Descartes came very close to beginning transcendental-philosophical investigations proper, and much more so than Kant. Descartes opened the way for a rigorous new philosophy but was held back by deep-seated preconceptions about the structure and method of this science (Husserl 1950: 63–64/23–25; 1954: 73ff./74ff.).

Husserl argues that Descartes' fundamental mistake was to assume that all theoretical and ideal sciences, and accordingly also philosophy, should adhere to or comply with the axiomatic-deductive model of the exact science of geometry. In the beginning of his *Cartesian Meditation*, Husserl argues:

Obviously it was, for Descartes, a truism from the start that the all-embracing science must have the form [*Gestalt*] of a deductive system, in which the whole structure rests, *ordine geometrico*, on an axiomatic foundation that grounds the deduction absolutely. For him a role similar to that of geometrical axioms in geometry is played in the all-embracing science by the axiom of the ego's absolute certainty of himself, along with the axiomatic principles innate in the ego – only this axiomatic foundation lies deeper than that of geometry and is called on to participate in the ultimate grounding even of geometrical knowledge. None of that shall determine our thinking. As beginning philosophers we do not as yet accept any normative ideal of science; and only so far as we produce one newly for ourselves can we ever have such an ideal. (Husserl 1950: 49/8–9)

For Husserl, the geometrical idea of science is merely a historically motivated set of value judgments and goals. He argues that, proceeding in a rigorous and truly radical manner, phenomenologists must suspend this set of judgments with all others and take as their task the illumination of its constitutive basis (Husserl 1954: 83–84/81–82, 268–269/264–265).

In Husserl's reading, Descartes' methodological preconceptions restricted the radicalism of his meditations. They suggested that *ego cogito* should, and could, function as a philosophical axiom from which metaphysical truths could be deduced as theorems. This prospect led Descartes to misunderstand the nature of the newly discovered ego and its activity of thinking.

Descartes took *ego cogito* as a self-evident truth from which he purported to derive his own existence as a “thinking thing” (*res cogitans*), as a part of the psychic or mental reality. In fact, Husserl argues, *ego cogito* cannot be used in this way. For radical doubt suspends not just the belief in the reality of extended things, but also the belief in the reality of thinking things. So, the thinking ego or self is not a thinking thing but irrecoverably distinct from all thingly being and all reality, both physical and psychic. On the other hand, it is still, after the suspension of belief, inseparably related to its object: the world and its multiple beings *as given in thinking*. This means that the phenomenological *epoché* or suspension does not nullify the world but gives the world to us in a new way. It opens a new transcendental area of correlative investigation, having two distinct but inseparable poles, the egoic acts of thinking or experiencing (*ego cogito*) and the objects thought or experienced (*cogitatum*) (Husserl 1950: 60–61/21).¹⁶ For Husserl, this is the proper subject matter of philosophy. In his reading, Descartes came close to realizing this objective but did not take the final decisive step of radically questioning the ideal model of science inherited from the Greeks (Husserl 1950: 63–64/24–25).

Thus, Husserl agrees with Descartes that philosophy can and must become a science, a “rigorous science,” as he calls it (Husserl [1911] 1965). But “rigor” does not mean exactitude in the geometrical or mathematical sense of the word. It means that the

¹⁶ In Husserl’s analysis, the ego is not a substantial entity, or an entity of any kind, real or constructed. It is a necessary structural element of all conscious experiences and of the continuous stream of experiencing. To highlight its non-substantial, non-thingly, and non-real character, Husserl calls it a “pole,” and argues that it stays constantly opposed to the other pole of experiencing, the objective one (*cogitatum*).

inquiry is purified from the preconceptions of everyday life as well as those of the positive sciences. The phenomenologist suspends his belief in the presence of practical and theoretical objects, and the world as the totality of all objects, and turns his attention to their ways of being given and then studies how the multiple senses of objectivity are constituted in conscious experiencing and in its passive undercurrents.

There is, however, a crucial analogy between geometry and phenomenology: both are ideal sciences that aim at describing ideal objects and necessary relations between such objects. What they thus share is an attitude toward particulars. The phenomenologist is interested in particular experiences in the same way as the geometrician is interested in particular triangles and circles. He does not collect large amounts of data and then make inductive generalizations based on them. He only goes through a few particulars and studies them as concrete examples of the essence of experience. His inquiry may be advanced more by a close inspection of a fictitious case than by a survey of actual instances (Husserl 1913: 160–163/181–183).

When the philosopher studies affective experience, for example, he is not interested in describing his own sentiments or the sentiments of his fellow men. This could be the task of the psychologist or the sociologist. The philosopher aims at finding the essential features that structure all – past, present, and future – modes of emotion. Further, his inquiry is not restricted to human emotions, but allows for the possibility of non-human affections, the emotions of animals, for example, or those of extraterrestrial aliens.

So, there is an important similarity between the objects studied in phenomenological philosophy and those of geometry. But the disparity is equally important. Husserl argues

that the ideal objects studied in phenomenology are very different in kind from the ideal objects studied in the mathematical sciences. Geometry deals with ideal space that cannot be seen or intuited but can only be extrapolated (Husserl [1939] 1985: 42/44). Its concepts are exact and they determine a closed complete whole. Phenomenology, on the other hand, deals with the essential structures of experiencing, and it discloses experience as through and through temporal. The continuum of experiences is not a closed manifold but has the character of a flow or a stream. Every singular experience is unique and irrecoverable; and together experiences form an open whole (Husserl 1952a: 130–131/242, 138–139/145; 1952b: 17, 44). Thus, phenomenology cannot form a system of founding axioms and deductive steps. It is not the “mathematics of experiences” but – by necessity – a purely descriptive science (Husserl 1913: 163–174/184–193).

To find the essential features of experience, the philosopher must compare particulars that are most different from one another. He cannot contend with what is familiar or common, but has to extend his study to exceptional cases and fictitious modifications. Husserl calls the process of looking for such examples, “free variation in imagination.” The philosopher starts from one particular experience and varies it in his imagination, trying to find the limit cases in which the experience transforms into some other mode of experiencing, from the perception of a human being to the perception of a dummy or an animal – and then further from perception to memory or imagination, from volition to emotion, from judgment to assumption and anticipation.

5. Further Perspectives

We have seen that there are important connections between Descartes' discussion of the mind-body union and classical phenomenology of perception and embodiment, as developed by Husserl and Merleau-Ponty. Further, the essay has also argued that the perplexing twofoldness of Descartes' account of the mind-body union – interactionistic on the one hand, and holistic on the other – can be clarified and made coherent by phenomenological analyses of the two different attitudes in which we can study human beings: naturalistic and personalistic.

In this perspective, the main guideline that Descartes offers to contemporary theorists of the human mind is the insight that our possibilities of understanding, explaining, and predicting the behaviors of human beings depend not merely on our advanced techniques of comparing and measuring the locations, states, and functions of their bodies or body parts, but also on our capacity to converse with them and immediately perceive or intuit their presence. Phenomenologists have argued that this immediate familiarity with other conscious beings cannot be derived from any results based on comparisons and measurements since it gives such beings, not as instances of universals, but as unique individuals. Moreover, classical phenomenologists have also maintained that in order to understand our relations to other conscious beings we must be clear about the nature of our own embodiment. The chapter explicated this philosophical heritage and ended in a discussion of the most important similarities and differences between Descartes' and Husserl's conceptions of philosophy as a radical science.

These are some of the main Cartesian ingredients that can be found in classical phenomenology. Of course, the reception of Descartes and Cartesianism by twentieth-century phenomenologists is not limited to these epistemological and methodological

results, and space here does not allow for a complete presentation of all the intricate debates on Descartes and Cartesianism that have evolved in phenomenology since its beginnings. However, some major tendencies and developments can, and need to be, pointed out to give a fuller picture of the reception.

In addition to Husserl's and Merleau-Ponty's constructive expositions of Cartesian insights, twentieth-century phenomenology also includes deeply critical reactions to Descartes and Cartesianism. The most influential of such approaches is Martin Heidegger's critique of all egological philosophy of consciousness and all subjectivism, starting with Descartes' own inquiries and including their radicalizations by Husserl and his epistemologically-minded followers. In *Being and Time* (*Sein und Zeit* 1927), Heidegger redefines phenomenology as a hermeneutical task of disclosing the primary facticity and temporality of our being-in-the-world. This, he argues, requires a historical dismantling of the universalistic concepts of subject, consciousness, ego, and person – as well as the related concepts of body and organism – handed down to us by our modern predecessors. The fundamental ontology outlined by Heidegger in *Being and Time* depends on the results of this critical undertaking.

In a series of historical-philosophical and phenomenological inquiries, Jean-Luc Marion (1987, 1986, cf. [1975] 1993) challenges the historical starting points of Heidegger's discussion of subjectivism and argues that the ego, discovered by Descartes and discussed by Husserl, must not be assimilated with an entity of any kind – psychological, social or metaphysical – but must be understood as a unique speaking position with a performative function and perlocutionary force (cf. Searle 1969; Austin 1975). The outcome of Marion's investigations is an original combination of Husserlian

and Heideggerian insights that aims at re-establishing phenomenology as an inquiry into pure givenness or manifestation, without any thesis for the primordially of the ego or that of Being.

There are also other alternative perspectives to Descartes' operative in phenomenology, both constructive and deconstructive. In *Totality and Infinity* (*Totalité et infini* 1961) and related works, Emmanuel Lévinas builds an ethical metaphysics that draws from Descartes' philosophy of infinities and magnitudes and from Kierkegaard's philosophy of faith. A Cartesian excursion, he claims, allows us to liberate our reflections from the dominant Kantian preconception about the primacy of finiteness in respect to infinity. In *Totality and Infinity*, we read:

For the Cartesian cogito is discovered, at the end of the Third Meditation, to be supported on the *certitude of the divine existence qua infinite*, by relation to which the finitude of the cogito (...) is posited and conceivable. This finitude could not be determined without the recourse to the infinite, as is the case in the moderns, for whom finitude is, for example, determined on the basis of the mortality of the subject. (...) This certitude [of consciousness] is due to the clarity and distinctness of the cogito, but certitude itself is sought because of the *presence of infinity* in this finite thought, which without this presence would be ignorant of its own finitude. (Lévinas [1961] 1971: xx/210, italics added)

Twentieth-century phenomenology also includes approaches that radically rework the intuition that led Descartes to construe his ontology as a dualism. The most influential of such dualistic approaches are Jean-Paul Sartre's existential phenomenology, which

operates by the distinction between being-in-itself and being-for-itself, explicated in his *Being and Nothingness* (*L'être et le néant* 1943), and Michel Henry's phenomenological Biranism, which operates by the concepts of exteriority and interiority, as worked out in his *The Essence of Manifestation* (*L'essence de la manifestation* 1963) and *Philosophy and Phenomenology of the Body* (*Philosophie et phénoménologie du corps* 1965).

Neither of these dualistic approaches establishes a dualism between mind and matter, or thought and extension. Rather, both work to redefine the concepts of subjectivity and objectivity, immanence and transcendence, in a way that corresponds to the richness of human experiencing.

All these different approaches share the core operative concepts of givenness and manifestation, and their main problematics concern the relations between the reflective self and what remains other or alien to it. In the debates between these phenomenological commentators, Descartes' work does not figure as a philosophical impasse but as a source of insight and motivation.

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